

Lossy Cavity

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Abstract

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Introduction

In the context of photoemissive sources and detection of emitted field, total system is separated in system S , which accounts for all internal workings of the source, and a reservoir R that carries the emitted fields. The reservoir describes the environment into which the sources losses energy.

Objective 1: discuss the master equation approach for theating a system constructing the emitted field in terms of the variables of the system S . Objective 2: construct the emitted field in terms of the variables of the system S

Derivation of master equation

Borh aproximation

We begin in with a Hamiltonian in a general form

$$H = H_S + H_R + H_{SR} \quad (1)$$

where H_S and H_R are Hamiltonian for S and R , respectively, and H_{SR} in and interaction Hamiltonian.

We wiil let $\chi(t)$ be the density operator for $S \otimes R$ and redine the redune density operator by $\rho(t)$ by

$$\rho = \text{tr}_R \chi \quad (2)$$

where the trace in only taken over the reservoir states.

In the interaction picture generated by the unitary transformation

$$U_c(t) = e^{H_c t / (i\hbar)}, \quad H_c = H_S + H_R \quad (3)$$

the density operador is written as

$$\tilde{\chi} = U_c^\dagger(t) \chi U_c(t) \quad (4)$$

The von Neumann equation in the interaction picture reads

$$\dot{\tilde{\chi}} = \frac{1}{i\hbar} [\tilde{H}_{SR}(t), \tilde{\chi}] \quad (5)$$

where

$$\tilde{H}_{SR}(t) = U_c^\dagger(t) H_{SR} U_c(t). \quad (6)$$

Integrating (5) and reinserting the solution in the same equation lead to

$$\begin{aligned} \dot{\tilde{\chi}}(t) = & \frac{1}{i\hbar} [\tilde{H}_{SR}(t), \tilde{\chi}(0)] \\ & - \frac{1}{\hbar^2} \int_0^t [\tilde{H}_{SR}(t), [\tilde{H}_{SR}(t'), \tilde{\chi}(t')]] \end{aligned} \quad (7)$$

We will assume the interaction is turned on at $t = 0$ and that no correlations existe between S and R at this initial time, then

$$\tilde{\chi}(0) = \chi(0) = \rho(0) R_0, \quad (8)$$

where R_0 in the initial reservoir density operatore.

Taking partial trace to (7) over reservoir states results in

$$\begin{aligned} \dot{\tilde{\rho}}(t) = & \frac{1}{i\hbar} [\text{tr}_R \{ \tilde{H}_{SR}(t) R_0 \}, \tilde{\rho}(0)] \\ & - \frac{1}{\hbar^2} \int_0^t \{ [\tilde{H}_{SR}(t), [\tilde{H}_{SR}(t'), \tilde{\chi}(t')]] \}. \end{aligned} \quad (9)$$

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The first term can be suppressed including $\text{tr}_R \{H_{SR}R_0\}$ in the system Hamiltonian. Consequently, operator coupling is set to have zero mean in the state R_0 .

Now, let us continue with two assumptions:

- The coupling is very weak, so for all times $\chi(t)$ should only show derivatino of order H_{SR} from an uncorrelated state.
- R is a large system whose state should be unaffected by its coupling to S .

Therefore

$$\tilde{\chi}(t) = \tilde{\rho}(t)R_0 + O(H_{SR}). \quad (10)$$

This expresión is known as *Borh approximation*.

It leds to

$$\dot{\tilde{\rho}}(t) = -\frac{1}{\hbar^2} \int_0^t \text{tr}_R \{ [\tilde{H}_{SR}(t), [\tilde{H}_{SR}(t'), \tilde{\rho}(t')R_0]] \}, \quad (11)$$

which is a Non-Markoffian equation because the future evolution of $\tilde{\rho}(t)$ depends on its past history through the integration over $\tilde{\rho}(t')$.

Markoff approximation

The reservoir is a large system maintained in thermal equilibrium, we do not expect it to preserve the minor chandes brought by its interaction with S for very long; not for long enough to signicantly affect the future evolution of S . It becomes a question of reservoir correlation time versus the time scale for significant change in S .

Let us treat a more specific model by writing

$$H_{SR} = \hbar \sum_i s_i \Gamma_i \quad (12)$$

The master equation in Born approximation is now

$$\begin{aligned} \dot{\tilde{\rho}}(t) = & -\frac{1}{\hbar^2} \int_0^t \sum_{ij} \\ & \left(\{ \tilde{s}_i(t) \tilde{s}_j(t') \tilde{\rho}(t') - \tilde{s}_j(t') \tilde{\rho}(t') \tilde{s}_i(t) \} \langle \Gamma_i(t) \Gamma_j(t') \rangle \right. \\ & \left. + \{ \tilde{\rho}(t') \tilde{s}_j(t') \tilde{s}_i(t) - \tilde{s}_i(t) \tilde{\rho}(t') \tilde{s}_j(t') \} \langle \Gamma_j(t') \Gamma_i(t) \rangle \right) \end{aligned} \quad (13)$$

where

$$\langle \Gamma_i(t) \Gamma_j(t') \rangle_R = \text{tr} \{ R_0 \Gamma_i(t) \Gamma_j(t') \} \quad (14)$$

$$\langle \Gamma_j(t') \Gamma_i(t) \rangle_R = \text{tr} \{ R_0 \Gamma_j(t') \Gamma_i(t) \} \quad (15)$$

are correlation functions.

If this correlation functions decay very rapidly on the time scale on which $\tilde{\rho}(t)$ varies, we might take

$$\langle \Gamma_i(t) \Gamma_j(t') \rangle \propto \delta(t - t'). \quad (16)$$

then we get Markoffian master equation. This is the *Markoff approximation*.

The Markoff approximation relies on the existence of two separated time scales: a low time scale for the dynamics of the system S , and a fast time scale characterizing the decay of reservoir correlation functions.

Master equation for a cavity mode driven by a thermal light

Let us consider and explicit model. Lets consider a ring cavity with reservoir comprised of travelling-wave modes that satisfy periodic boundary condiciones at $z = -L/2$ and $z = L/2$. The chosen cavity mode couples to the reservoir throught a partially transmitting mirror.

The free and interaction Hamiltonians are the following

$$H_S = \hbar \omega_c a^\dagger a \quad (17)$$

$$H_R = \sum_j \hbar \omega_j r_j^\dagger r_j \quad (18)$$

$$H_{SR} = \sum_j k_j^* a r_j^\dagger + k_j a^\dagger r_j = a \Gamma^\dagger + a^\dagger \Gamma \quad (19)$$

System S is an harmonic oscillator of frequency ω_c while reservoir is a collection of harmonic oscillators with frecuencies ω_j . System and reservoir modes are coupled by coupling constant k_j in the rotating wave approximation.

We take the reservoir to be in thermal equilibruim at temperatur T :

$$R_0 = \bigotimes_j e^{-\hbar \omega_j r_j^\dagger r_j / k_B T} (1 - e^{-\hbar \omega_j / k_B T}). \quad (20)$$

Identification with (12) is made by setting

$$s_1 = a \quad s_2 = a^\dagger \quad (21)$$

$$\Gamma_1 = \Gamma^\dagger \quad \Gamma_2 = \Gamma \quad (22)$$

Then master equation (13) is written as

$$\begin{aligned} \dot{\tilde{\rho}}(t) = & -\frac{1}{\hbar^2} \int_0^t \\ & \left\{ [aa\tilde{\rho}(t') - a\tilde{\rho}(t')a] e^{-i\omega_c(t-t')} \langle \Gamma^\dagger(t) \Gamma^\dagger(t') \rangle_R + h.c. \right. \\ & + [a^\dagger a^\dagger \tilde{\rho}(t') - a^\dagger \tilde{\rho}(t') a^\dagger] e^{i\omega_c(t-t')} \langle \Gamma(t) \Gamma(t') \rangle_R + h.c. \\ & + [aa^\dagger \tilde{\rho}(t') - a^\dagger \tilde{\rho}(t') a] e^{-i\omega_c(t-t')} \langle \Gamma^\dagger(t) \Gamma(t') \rangle_R + h.c. \\ & \left. + [a^\dagger a \tilde{\rho}(t') - a \tilde{\rho}(t') a^\dagger] e^{i\omega_c(t-t')} \langle \Gamma(t) \Gamma^\dagger(t') \rangle_R + h.c. \right\} \end{aligned} \quad (23)$$

where the reservoir correlation functions are

$$\langle \tilde{\Gamma}^\dagger(t) \tilde{\Gamma}^\dagger(t') \rangle_R = 0 \quad (24)$$

$$\langle \tilde{\Gamma}(t) \tilde{\Gamma}(t') \rangle_R = 0 \quad (25)$$

$$\langle \tilde{\Gamma}^\dagger(t) \tilde{\Gamma}(t') \rangle_R = \sum_i |k_i|^2 e^{i\omega_i(t-t')} \bar{n}(\omega_i, T) \quad (26)$$

$$\langle \tilde{\Gamma}(t) \tilde{\Gamma}^\dagger(t') \rangle_R = \sum_i |k_i|^2 e^{-i\omega_i(t-t')} [\bar{n}(\omega_i, T) + 1] \quad (27)$$

with

$$\bar{n}(\omega_i, T) = \frac{1}{(1 - e^{-\hbar\omega_i/k_B T})}. \quad (28)$$

We will change this summation to an integration by introducing a density of states $g(\omega)$, such that $g(\omega)d\omega$ gives the number of oscillators with frequencies in the interval ω to $\omega + d\omega$.

Making the change of variable $\tau = t - t'$ in (23), this equation can be restated as

$$\begin{aligned} \dot{\rho} = & - \int_0^t \left\{ \right. \\ & + \left[a a^\dagger \tilde{\rho}(t-\tau) - a^\dagger \tilde{\rho}(t-\tau) a \right] e^{-i\omega_c \tau} \langle \Gamma^\dagger(t) \Gamma(t-\tau) \rangle_R \\ & + h.c. \\ & + \left[a^\dagger a \tilde{\rho}(t-\tau) - a \tilde{\rho}(t-\tau) a^\dagger \right] e^{i\omega_c \tau} \langle \Gamma(t) \Gamma^\dagger(t-\tau) \rangle_R \\ & \left. + h.c. \right\} \end{aligned} \quad (29)$$

where the non-zero correlation functions are

$$\begin{aligned} \langle \tilde{\Gamma}^\dagger(t) \tilde{\Gamma}(t-\tau) \rangle_R &= \int d\omega e^{i\omega\tau} g(\omega) |\kappa(\omega)|^2 \bar{n}(\omega, T) \\ \langle \tilde{\Gamma}(t) \tilde{\Gamma}^\dagger(t-\tau) \rangle_R &= \int d\omega e^{i\omega\tau} g(\omega) |\kappa(\omega)|^2 [\bar{n}(\omega, T) + 1] \end{aligned} \quad (30)$$

The variable τ measures how far back the reservoir is remembering.

We can certainly see that for τ large enough the oscillation exponential will average the slowly varying functions $g(\omega)$, $|\kappa(\omega)|^2$ and $\bar{n}(\omega, t)$ essentially to zero.

Time scales estimation

Time scale of system S dynamics The free oscillation at the frequency ω_c is removed by the transformation to interaction picture; therefore, we expect that the remaining time dependence in $\tilde{\rho}$ is characterized by a decay time for the cavity mode by inverse of the mode linewidth, this with typically be a number of the order of

$$\tau_S \sim 10^{-8} \text{sec}. \quad (31)$$

Time scale for decay of correlation functions To estimate the reservoir correlation time let us just take $\kappa(\omega)$ and $g(\omega)$ as constants and focus on frequency dependence of $\bar{n}(\omega, T)$. Because of $e^{\pm i\omega_c t}$ multiplying the reservoir correlation function, it is really only the $\omega \approx \omega_c$ part of frequency range in the integral that is important. We can estimate the correlation time through the inverse on spectral width of $\bar{n}(|\omega|, T)$:

$$\tau_R = \frac{1}{\Delta\omega_R} \sim \frac{\hbar}{k_B T}. \quad (32)$$

At room temperature this gives a number of order of

$$\tau_R \sim 0.25 \times 10^{-13} \text{sec}. \quad (33)$$

Markoff approximation

Now we are satisfied that the τ integration in (29) is dominated by times that are much shorter than the time scale for the evolution of $\tilde{\rho}$, we replace $\tilde{\rho}(t-\tau)$ by $\tilde{\rho}(t)$ and obtain

$$\begin{aligned} \dot{\rho} = & \alpha(a\tilde{\rho}a^\dagger - a^\dagger a\tilde{\rho}) + h.c. \\ & + \beta(a\tilde{\rho}a^\dagger + a^\dagger \tilde{\rho}a - a^\dagger a\tilde{\rho} - \tilde{\rho}a^\dagger a)h.c. \end{aligned} \quad (34)$$

with

$$\alpha = \int_0^t d\tau \int_0^\infty d\omega e^{-i(\omega-\omega_c)\tau} g(\omega) |\kappa(\omega)|^2, \quad (35)$$

$$\beta = \int_0^t d\tau \int_0^\infty d\omega e^{-i(\omega-\omega_c)\tau} g(\omega) |\kappa(\omega)|^2 \bar{n}(\omega, T). \quad (36)$$

Since τ integration is dominated by much shorter times than time scale for changes in $\tilde{\rho}$, we can extend the τ integration to infinity and evaluate α and β using

$$\lim_{t \rightarrow \infty} \int_0^t d\tau e^{-i(\omega-\omega_c)\tau} = \pi \delta(\omega - \omega_c) + i \frac{P}{\omega_c - \omega}, \quad (37)$$

where P indicates the Cauchy principal value. We find

$$\alpha = \pi g(\omega_c) |\kappa(\omega_c)|^2 + i\Delta \quad (38)$$

$$\beta = \pi g(\omega_c) |\kappa(\omega_c)|^2 \bar{n}(\omega_c, T) + i\Delta'. \quad (39)$$

with

$$\Delta = P \int_0^\infty d\omega \frac{g(\omega) |\kappa(\omega)|}{\omega_c - \omega} \quad (40)$$

$$\Delta' = P \int_0^\infty d\omega \frac{g(\omega) |\kappa(\omega)|}{\omega_c - \omega} \bar{n}(\omega, T). \quad (41)$$

Then

$$\begin{aligned} \dot{\rho} = & -i\Delta [a^\dagger a, \rho] + \kappa(2a\tilde{\rho}a^\dagger - a^\dagger a\tilde{\rho} - \tilde{\rho}a^\dagger a) \\ & + 2\kappa\bar{n}(a\tilde{\rho}a^\dagger + a^\dagger \tilde{\rho}a - a^\dagger a\tilde{\rho} - \tilde{\rho}aa^\dagger), \end{aligned} \quad (42)$$

where

$$\kappa = \pi g(\omega_C) |\kappa(\omega_C)|^2, \quad \bar{n} = \bar{n}(\omega_C, T). \quad (43)$$

Returning to Schrödinger picture using

$$\dot{\rho} = \frac{1}{i\hbar} [H_S, \rho] + e^{-(i/\hbar)H_S t} \dot{\tilde{\rho}} e^{(i/\hbar)H_S t} \quad (44)$$

we obtain the *master equation for a cavity mode driven by thermal light*.

$$\begin{aligned} \dot{\rho} = & -i\omega'_C [a^\dagger a, \rho] + \kappa(2a\rho a^\dagger - a^\dagger a\rho - \rho a^\dagger a) \\ & + 2\kappa\bar{n}(a\rho a^\dagger + a^\dagger \rho a - a^\dagger a\rho - \rho a a^\dagger), \end{aligned} \quad (45)$$

where

$$\omega'_C = \omega_C + \Delta. \quad (46)$$

In modern notation:

$$\mathcal{D}[a]\rho = a\rho a^\dagger - 1/2\{a^\dagger a, \rho\} \quad (47)$$

$$\mathcal{D}[a^\dagger]\rho = a^\dagger \rho a - 1/2\{a a^\dagger, \rho\} \quad (48)$$

then

$$\dot{\rho} = -i\omega'_C [a^\dagger a, \rho] + 2\kappa(\bar{n} + 1)\mathcal{D}[a]\rho + 2\kappa\bar{n}\mathcal{D}[a^\dagger]\rho. \quad (49)$$

Cavity output